1 Two-stage ER-DRO model (CVaR-type ambiguity set)

Data: (x^k, p^k), $k = 1, \ldots, N$; for example, the context x^k (zero-vehicle household share) and the robotaxi choice probability p^k observed per city–period in which robotaxis have operated. A regression yields $\hat{f}$ and the logit-scale residuals $\hat{\varepsilon}^k = \mathrm{logit}(p^k) - (\hat{\beta}_0 + \hat{\beta}_1 x^k)$ (§1.3); attaching one residual at a time to the context x_c of candidate city c generates the scenarios D_c^s. Since one residual corresponds to one scenario, the scenario set S is the index set of the residuals and $|S| = N$. Below, the model is written as the first-stage problem (1), the per-scenario second-stage problem (3), and the extensive form (6) that is actually solved. The first-stage variables are (y, q), and the second-stage variables (O^s, U^s) are separable by scenario s. No separate term is kept for the per-vehicle deployment/operating cost. Since every term of the model is expressed in trip units, the per-vehicle cost K_c is converted into a cost per unit of capacity K_c/κ_c and absorbed into c_c^+, c_c^- (§1.6); keeping it as an independent term does not change the optimal solution.

1.1 First-stage problem

$$\min_{y,q} \quad \underbrace{\sum_{c\in C} F_c y_c}_{\text{entry cost}} + \underbrace{\mathcal{R}_\zeta(q)}_{\text{CVaR of scenario penalties}} \tag{1a}$$

$$\text{s.t.} \quad 0 \le q_c \le \bar{q}_c\, y_c, \qquad \forall c \in C, \tag{1b}$$

$$\sum_{c\in C} q_c \le Q, \tag{1c}$$

$$y_c \in \{0,1\},\ q_c \in \mathbb{Z}_{\ge 0}, \qquad \forall c \in C. \tag{1d}$$

Here $\mathcal{R}_\zeta(q)$ is the CVaR of the second-stage cost $H(q, D^s)$ over the scenario distribution, written in the Rockafellar–Uryasev threshold form

$$\mathcal{R}_\zeta(q) = \min_{\tau\in\mathbb{R}} \Big\{ \tau + \frac{1}{|S|(1-\zeta)} \sum_{s\in S} \big(H(q, D^s) - \tau\big)^+ \Big\}. \tag{2}$$

When $\zeta = 0$, $\mathcal{R}_0(q) = \frac{1}{|S|}\sum_s H(q, D^s)$ and the model reduces to ER-SAA (sample average). The context x_c enters only through the scenario values $D_c^s = M_c p_c^s$; the optimization structure is unchanged.

1.2 Second-stage problem (per-scenario recourse)

Given the first-stage decision q and the demand $D^s = (D_c^s)_{c\in C}$ of scenario s, the second-stage cost is

$$H(q, D^s) = \min_{O^s, U^s} \quad \sum_{c\in C} \big(c_c^+ O_c^s + c_c^- U_c^s\big) \tag{3a}$$

$$\text{s.t.} \quad \kappa_c q_c - D_c^s = O_c^s - U_c^s, \qquad \forall c \in C, \tag{3b}$$

$$O_c^s,\ U_c^s \ge 0, \qquad \forall c \in C. \tag{3c}$$

(3) is also separable by c, and its optimal solution has the closed form $O_c^s = (\kappa_c q_c - D_c^s)^+$, $U_c^s = (D_c^s - \kappa_c q_c)^+$. Hence $H(q, D^s) = \sum_c \left[c_c^+ (\kappa_c q_c - D_c^s)^+ + c_c^- (D_c^s - \kappa_c q_c)^+\right]$ is a convex piecewise-linear function of q, which is the basis for applying two-stage decomposition (L-shaped). This second-stage cost has the same structure as the overage/underage cost of the newsvendor problem, and the first-stage decision q is linked to the second stage only through the CVaR in (2).

1.3 Demand model: $D_c^s = M_c\, p_c^s$ and the estimation of p

The robotaxi demand of city c is the product $D_c = M_c\, p_c$ of the potential trip opportunities M_c and the choice probability p_c, and uncertainty is placed only in p_c. M_c is a fixed input, and p_c is estimated by a logit model, i.e. a function of the exogenous context x_c (ACS zero-vehicle household share) plus an error term.

$$\mathrm{logit}(p) = \ln\frac{p}{1-p} = \beta_0 + \beta_1 x + \varepsilon, \qquad p = \sigma(\beta_0 + \beta_1 x + \varepsilon), \quad \sigma(z) = \frac{1}{1+e^{-z}}, \quad \mathbb{E}[\varepsilon] = 0. \tag{4}$$

Using observations (x^k, p^k), $k = 1, \dots, N$ from places where robotaxis have operated, $z^k = \mathrm{logit}(p^k)$ is fitted on x^k by least squares to obtain $\hat{\beta}_0, \hat{\beta}_1$ and the residuals $\hat{\varepsilon}^k = z^k - \hat{\beta}_0 - \hat{\beta}_1 x^k$; for candidate city c the residuals are attached one at a time to form scenarios.

$$p_c^s = \sigma\big(\hat{\beta}_0 + \hat{\beta}_1 x_c + \hat{\varepsilon}^s\big), \qquad D_c^s = M_c\, p_c^s, \qquad s = 1, \dots, N. \tag{5}$$

Since x_c is an exogenous variable independent of the decision (y_c, q_c), all D_c^s are constants computed before optimization, and the structure of (1)–(6) is unchanged. The share p rather than the demand D is regressed because an error on the D scale is proportional to M_c and therefore heteroscedastic across city sizes, so the residuals could not be pooled into one sample.

There are two reasons for the logit form. First, p is "the probability that one trip opportunity chooses a robotaxi", so the standard way to write a choice probability as a function of explanatory variables is the discrete-choice logit, which automatically guarantees $p \in (0, 1)$ and monotonicity in x. Second, Loa et al. (2026), the only study analyzing observed robotaxi demand, report the zero-vehicle household share as a positive and significant variable in a log-linear spatial lag model with $\ln(\text{trips} + 1)$ as the dependent variable; with $D = Mp$, a log-linear trip model has the same form as $\mathrm{logit}(p) \approx \ln p$ when p is small, so (4) is consistent with that result. This is also the basis for using a single context, the zero-vehicle household share.

1.4 Extensive form: the MILP actually solved

Linearizing $(\cdot)^+$ in (2) with variables w^s and merging (3) into (1) gives the following single MILP. (6b)–(6c), (6g)–(6h) are second-stage constraints separable by scenario, and (6d)–(6f) are first-stage constraints.

$$\min_{y,q,O,U,\tau,w} \quad \sum_{c\in C} F_c y_c + \tau + \frac{1}{|S|(1-\zeta)} \sum_{s\in S} w^s \tag{6a}$$

$$\text{s.t.} \quad w^s \geq \sum_{c\in C} \big(c_c^+ O_c^s + c_c^- U_c^s\big) - \tau, \qquad \forall s \in S, \tag{6b}$$

$$\kappa_c q_c - D_c^s = O_c^s - U_c^s, \qquad \forall c \in C,\ s \in S, \tag{6c}$$

$$0 \leq q_c \leq \bar{q}_c\, y_c, \qquad \forall c \in C, \tag{6d}$$

$$\sum_{c\in C} q_c \leq Q, \tag{6e}$$

$$y_c \in \{0, 1\},\ q_c \in \mathbb{Z}_{\geq 0}, \qquad \forall c \in C, \tag{6f}$$

$$O_c^s,\ U_c^s \geq 0, \qquad \forall c \in C,\ s \in S, \tag{6g}$$

$$w^s \geq 0, \quad \tau \in \mathbb{R}, \qquad \forall s \in S. \tag{6h}$$

1.5 Variant: entry cost as a budget constraint

If the number of cities that can be entered (or the entry budget) is the real constraint, remove $\sum_c F_c y_c$ from the objective (6a) and add the following constraint.

$$\sum_{c\in C} F_c y_c \leq B \qquad \text{or} \qquad \sum_{c\in C} y_c \leq B. \tag{7}$$

In this variant only $\tau + \frac{1}{|S|(1-\zeta)} \sum_s w^s$, i.e. the CVaR of the second-stage penalty, remains in the objective. Because the variant directly restricts the combinations of y, it mitigates the problem in two-stage decomposition that the initial master problem returns the trivial solution $(y = q = 0)$ and the iterations become long.

1.6 Absorption of the per-vehicle cost: why the K_c term is dropped

Even if the per-vehicle cost K_c (deployment/operation) is placed in the objective as $\sum_c K_c q_c$, by (6c) we have $\kappa_c q_c = D_c^s + O_c^s - U_c^s$ for every s, so for $\zeta = 0$

$$\sum_c K_c q_c + \frac{1}{|S|} \sum_s \sum_c \big(c_c^+ O_c^s + c_c^- U_c^s\big) = \underbrace{\frac{1}{|S|} \sum_s \sum_c \frac{K_c}{\kappa_c} D_c^s}_{\text{constant independent of the decision}} + \frac{1}{|S|} \sum_s \sum_c \left[\left(c_c^+ + \frac{K_c}{\kappa_c}\right) O_c^s + \left(c_c^- - \frac{K_c}{\kappa_c}\right) U_c^s\right] \tag{8}$$

holds. That is, K_c is equivalent to the redefinition $c_c^+ \leftarrow c_c^+ + K_c/\kappa_c$, $c_c^- \leftarrow c_c^- - K_c/\kappa_c$ (for $\zeta > 0$ the term $(K_c/\kappa_c)D_c^s$ enters the CVaR, so this is an approximation), and it is not an independent decision lever. Moreover, if K_c is the same across cities and (6e) binds with equality, $\sum_c K_c q_c = KQ$ is a constant. Therefore this study does not keep K_c as a separate term; the per-vehicle cost converted to a per-trip-of-capacity value is regarded as included in the definitions of c_c^+, c_c^-, and only their ratio is considered (e.g. 1:5).

Constraint	Stage	Meaning
$0 \le q_c \le \bar{q}_c y_c$ (6d)	stage 1	no vehicles unless the city is entered; if entered, up to the per-city cap
$\sum_c q_c \le Q$ (6e)	stage 1	limit on the total number of vehicles
$\sum_c F_c y_c \le B$ (7)	stage 1 (variant)	limit on the entry budget (or number of entered cities); replaces $\sum_c F_c y_c$ in the objective
$\kappa_c q_c - D_c^s = O_c^s - U_c^s$ (6c)	stage 2 (separable by s)	service capacity minus robotaxi demand = surplus capacity − unmet demand
$w^s \ge \sum_c (c_c^+ O_c^s + c_c^- U_c^s) - \tau$, $w^s \ge 0$ (6b),(6h)	stage 2 (separable by s)	amount by which the penalty of scenario s exceeds the threshold τ; in the minimization $w^s = (\text{penalty}_s - \tau)^+$

2 Notation

2.1 Additional notation for the contextual and DRO parts

Symbol	Meaning	Type and unit
(x^k, p^k)	context (zero-vehicle household share) and robotaxi choice probability of observation k	data; $k = 1, \dots, N$
x_c	covariate vector of candidate city c; at this stage a single variable, the zero-vehicle household share	input; corresponds to the new observation x in Kannan et al.
$\hat{f}$	estimated regression function; $\hat{f}(x) \approx \mathbb{E}[p \mid x]$. Logit $\hat{f}(x) = \sigma(\hat{\beta}_0 + \hat{\beta}_1 x)$, $\sigma(z) = 1/(1+e^{-z})$	estimated function
$\hat{\varepsilon}^k$	residual $\text{logit}(p^k) - (\hat{\beta}_0 + \hat{\beta}_1 x^k)$	derived; logit scale
ζ	radius (tail fraction) of the CVaR-type ambiguity set; $1-\zeta$ is the fraction of scenarios kept as the worst group	input; $[0, 1)$; $\zeta = 0$ gives ER-SAA
τ	penalty threshold; a decision variable equal at the optimum to "the $\lvert S\rvert(1-\zeta)$-th largest penalty"	decision variable; monetary unit
w^s	amount by which the penalty of scenario s exceeds τ	decision variable; monetary unit
$\mathcal{R}_\zeta(q)$	CVaR of the second-stage cost, eq. (2)	function of q; monetary unit
$H(q, D^s)$	second-stage (recourse) cost of scenario s, eq. (3)	function of q; monetary unit
B	entry budget or number of cities that can be entered (variant)	input; monetary unit or number of cities

2.2 Basic notation

Symbol	Meaning	Type and unit
C	set of candidate cities	input set; cities
S	set of demand scenarios	input set; index set of residuals, $\lvert S\rvert = N$
y_c	1 if city c is entered, 0 otherwise	first-stage decision; binary
q_c	number of vehicles deployed in city c	first-stage decision; vehicles (integer)
κ_c	trips one vehicle can serve during the evaluation period	scale parameter; trips/vehicle
Q	total number of vehicles available for deployment (owned fleet; acquisition cost and depreciation are sunk)	input; vehicles
$\bar{q}_c$	per-city deployment cap	input; vehicles
F_c	fixed cost of entry and basic infrastructure	input; monetary unit per evaluation period
c_c^+	loss per unit of surplus capacity	input; money/trip
c_c^-	opportunity loss per unit of unmet potential demand; eq. (8)	input; money/trip
M_c	annual number of **potential trip opportunities over all modes** in city c	input; opportunities; data or assumption
p_c^s	probability that a trip opportunity in city c chooses a robotaxi	generated by regression; $[0, 1]$; $p_c^s = \sigma(\hat{\beta}_0 + \hat{\beta}_1 x_c + \hat{\varepsilon}^s)$
D_c^s	robotaxi demand	derived input; trips; $D_c^s = M_c p_c^s$

Symbol	Meaning	Type and unit
O_c^s	surplus service capacity	second-stage outcome; trips; $O_c^s = (\kappa_c q_c - D_c^s)^+$
U_c^s	unmet robotaxi demand	second-stage outcome; trips; $U_c^s = (D_c^s - \kappa_c q_c)^+$

3 Numerical experiments: synthetic data

3.1 Overall flow

In a synthetic environment where the true demand model is known, we measure (i) whether the estimation–optimization pipeline works, and (ii) quantities that can only be measured when the true model is available (out-of-sample cost, value of context, effect of DRO). The flow has five steps.

1. Set the candidate-city inputs $(x_c, M_c, \kappa_c, \bar{q}_c, F_c)$ and the true demand model $p = \sigma(\beta_0 + \beta_1 x + \varepsilon)$.
2. Generate training data $(x^k, p^k)_{k=1}^N$ from the true model and estimate $\hat{f}$ and the residuals $\hat{\varepsilon}^k$ by logit OLS.
3. Build the per-city scenarios $D_c^s = M_c\, \sigma(\hat{\beta}_0 + \hat{\beta}_1 x_c + \hat{\varepsilon}^s)$ and solve the extensive form (6).
4. Compute the out-of-sample expected cost and tail cost of the solution (y, q) on 1,000 evaluation scenarios generated separately from the true model.
5. Repeat 2–4 over the settings (N, c^-, ζ) and compare with the no-context benchmark and the variant (7).

3.2 Synthetic data generation

Candidate-city inputs. The candidate cities are the 18 cities Waymo has announced as future expansion candidates. Each city is mapped to one central county (e.g. Chicago → Cook County), and the context x_c is the actual zero-vehicle household share from ACS 2019–2023 5-year estimates, table B08201 (range 0.054–0.777; New York County at 0.777 is an extreme value covering Manhattan only). The potential trip opportunities M_c are the total households in the same ACS table multiplied by a proportionality constant of 10; this constant is a toy assumption (with real data it is replaced by the NHTS × ACS cross-classification). The remaining inputs are common to all cities (Table 4).

True demand model. The true model is the logit model (4) of §1.3 with $\varepsilon \sim N(0, \sigma_\varepsilon^2)$.

$$\operatorname{logit}(p) = \beta_0 + \beta_1 x + \varepsilon, \qquad \varepsilon \sim N(0, \sigma_\varepsilon^2). \tag{9}$$

$\beta_1 = 0.8$ is the zero-vehicle household share coefficient 0.008 per %-point ($t = 7.4$) in Table 3 of Loa et al. (2026) rescaled to the 0–1 scale (0.008×100). $\beta_0 = -3.5$ and $\sigma_\varepsilon = 0.3$ are assumptions chosen so that p falls roughly in the range 0.03–0.09.

Base parameters. Costs are normalized with $c^+ = 1$ and c^- is varied over $1, 5, 10$ (to examine the effect of the overage-to-underage cost ratio $c^+ : c^-$ on the decision). The entry cost is set as $F_c = F_0\, c^-$ so that the "entry cost to unmet margin" ratio stays independent of c^-.

Table 4: Base parameters and rationale

Parameter	Value	Rationale
x_c	actual ACS values (0.054–0.777)	ACS 2019–2023 B08201, county level
M_c	total households × 10	household counts are actual ACS values; the constant is a toy assumption
κ_c	8,760 trips/vehicle/year	24 trips/day × 365 days (sensitivity 6570–10950); 24 trips per vehicle per day back-calculated from Waymo's public figures
$\bar{q}_c$	150 vehicles	assumption (sensitivity 50–300)
Q	$0.6 \times \mathbb{E}[\sum_c D_c]/\kappa$	assumption; only 60% of expected demand can be served, so that the allocation is non-trivial
F_0	10^5 (at $c^- = 1$)	assumption (sensitivity 0.25×10^5–4×10^5)
$c^+ : c^-$	$1:1,\ 1:5,\ 1:10$	assumption
$\beta_0, \beta_1, \sigma_\varepsilon$	$-3.5,\ 0.8,\ 0.3$	β_1 rescaled from Loa et al.; β_0, σ_ε assumed
N	15, 200	scarce/ample city–period observations; number of training observations = number of scenarios $\lvert S\rvert$
ζ	0, 0.5, 0.9	0 = ER-SAA; 0.9 = default
Evaluation scenarios	1,000	arbitrary assumption

3.3 Training and evaluation data

Training data. These mimic "(x, p) observed in city–periods where robotaxis have already operated". Random x^k are drawn uniformly over the range of the candidate-city x_c, ε^k is drawn from the true model (9), and p^k is computed ($k = 1, \ldots, N$). Without knowing the true coefficients, the model estimates $\hat{\beta}_0, \hat{\beta}_1$ and the residuals $\hat{\varepsilon}^k$ from these N points, and the optimization sees only the scenarios built from those estimates. With real data this step is replaced by an observed table with x = ACS share and p = robotaxi trips/M.

Evaluation data. These mimic "the demand that actually materializes after the decision is made". The true coefficients are applied to the actual x_c of the candidate cities, and 1,000 draws of ε independent of the training data produce D_c^{eval} (18 candidate cities × 1,000). The optimization never sees these data; once (y, q) is fixed, the per-scenario cost is computed in the closed form of eq. (3) and scored by the mean (expected cost) and the mean of the top 10% (tail cost, $\text{CVaR}_{0.9}$). Because the evaluation data are fixed independently of N, the decisions made with $N = 15$ and $N = 200$ are compared on the same yardstick. The value scored on the training data (in-sample) is optimistic because the optimization was fitted to those data, which is why in-sample and OOS differ in Table 7.

Solution method comparison: extensive form vs. L-shaped decomposition. The two-stage model (1)–(3) was solved (i) as a whole in the extensive form (6) and (ii) by multi-cut L-shaped decomposition into a master (y, q, τ) and per-scenario subproblems ($N = 15$, $c^- = 5$, $\zeta = 0.9$; the decomposition stops at $\text{UB} - \text{LB} \leq 10^{-2}\,|\text{UB}|$, i.e. a 1% relative tolerance).

Table 5: Extensive form vs. L-shaped decomposition

Setting	Method	Objective	Time	Note
$B = 3$	extensive	16,036,515	0.4 s	entered: Chicago, New York, San Diego
	L-shaped	16,140,861	29.5 s	38 iterations, same cities, difference 0.65%
no B	extensive	17,304,198	1.5 s	entered: Chicago, New York, Philadelphia, Seattle
	L-shaped	17,378,353	54.5 s	38 iterations, difference 0.43%, 2 cities differ

The L-shaped objective was 0.65% higher than the extensive form for $B = 3$ and 0.43% higher without B; the entered cities were identical for $B = 3$ but two differed without B. Since the value returned by L-shaped is the cost of the feasible solution at termination (an upper bound), the difference is the result of stopping early at the tolerance, and it shows that without B a different city combination is feasible at a cost 0.43% higher. Computation times were 0.4 s and 1.5 s for the extensive form versus 29.5 s and 54.5 s for L-shaped, so the extensive form was far faster, because the second-stage cost is the max of two affine functions and the O, U variables of the extensive form amount to including all cuts from the start. Moving the $F_c y_c$ term into the

constraint B ($\sum_{c\in C} y_c \le B$) reduces the convergence time of the decomposition. All subsequent experiments use the extensive form; the comparison will be revisited for the decision-dependent extension.

3.4 Analysis of results

The experimental setting is as in §3.2, and the total fleet computed from the true model is $Q = 267$ vehicles. Every solution is evaluated out of sample on 1,000 true-model scenarios (expected cost and tail cost = mean of the worst 10%). The five questions below are each answered in the order answer, evidence, and the corresponding table/figure.

Summary. (Q1) Logit regression of p recovers the true coefficients and satisfies the residual-pooling condition. (Q2) The conclusions are insensitive to the $c^+ : c^-$ ratio, so $1 : 5$ is taken as the base. (Q3) Context reduces cost in proportion to how strongly x drives demand (1–3% in the current setting, 25–41% with strong context); DRO gives up expected cost to reduce tail cost, and the two should be used together. (Q4) Even with 15 observations the cost stays within 3%. (Q5) Cost is sensitive to the scale of Q, κ, F_0, σ_ε; Chicago and New York are selected in every setting and San Diego in 17 of 18.

Q1. Can demand be estimated by a logit regression of p? In Table 6, with $N = 200$, $\hat\beta_1 = 0.67$ and $\hat\sigma_\varepsilon = 0.30$ recover values close to the truth (0.8, 0.3). The condition for pooling the residuals as scenarios (mean 0 and constant variance regardless of x) is confirmed by the residual scatter in Figure 1. The low test R^2 of 0.12–0.14 arises because, by construction, x explains only about 13% of the demand variation; using only the mean without x gives $R^2 \approx 0$, so the explanatory power of x is small but clearly present.

Table 6: Logit OLS estimates (error metrics on the p scale, 1,000 test observations)

N	$\hat\beta_0$	$\hat\beta_1$	$\hat\sigma_\varepsilon$	test MSE	test MAE	test ME	test R^2	no-context R^2
15	−3.655	1.114	0.325	0.0002	0.0101	0.0461	0.124	−0.011
200	−3.479	0.665	0.298	0.0002	0.0099	0.0479	0.143	−0.013
true	−3.500	0.800	0.300					

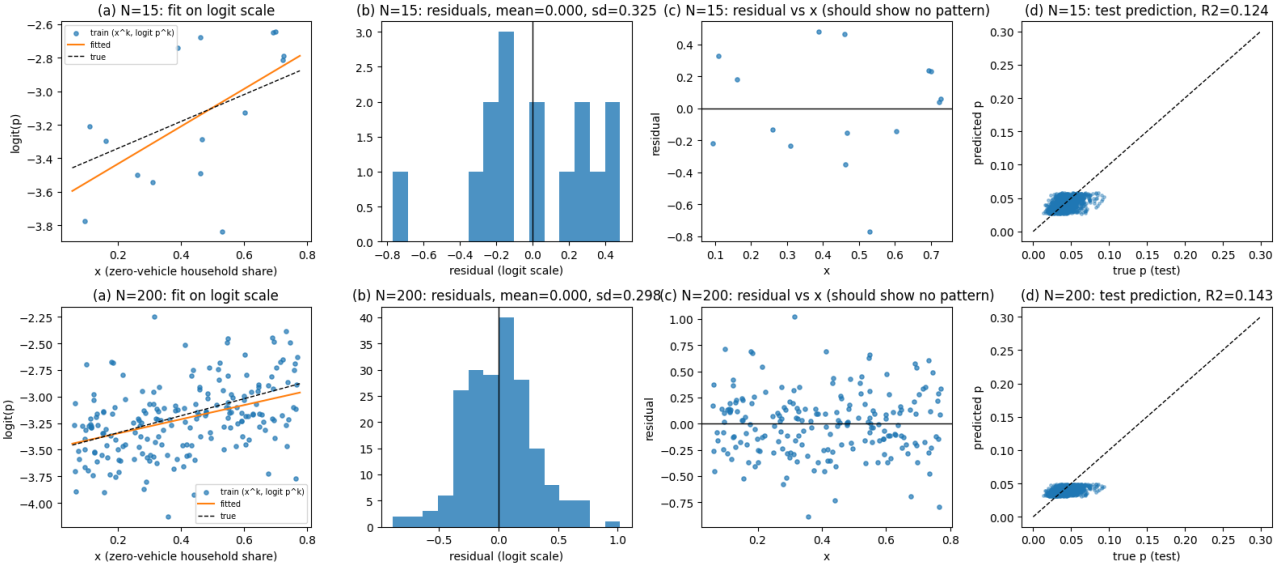


Figure 1: Diagnostics of the estimate of f. (a) data, fitted line and true line on the logit scale, (b) residual distribution, (c) residual vs. x, (d) predicted vs. true p on the test data. Top row $N = 15$, bottom row $N = 200$.

Q2. Which $c^+ : c^-$ ratio should be used? The ratio is an input chosen in the absence of data, and the role of the experiment is to show that the conclusions do not change with it. In Table 7, for all of $c^- = 1, 5, 10$: (i) the context model has lower OOS expected cost than the no-context model, (ii) the number of entered cities falls as ζ grows, and (iii) Chicago and New York are selected in all 18 settings and San Diego in 17. In the c^-

panel of Figure 3, cost grows in proportion to c^- (3.08, 5.79, 14.22, 27.63, 54.45M at $c^- = 1, 2, 5, 10, 20$) but the relative order of the three models does not change. Therefore the base setting is $1 : 5$, where underage is costlier than overage, and the other ratios are only reported.

Table 7: Optimal number of entered cities, in-sample objective, OOS cost ($c^+ = 1$ units) and context gain by setting

N	c^-	ζ	# entered	in-sample	OOS mean	OOS tail	no-ctx mean	no-ctx tail	gain
15	1	0	9	2,490,926	2,695,476	4,857,473	2,758,361	4,657,473	2%
		0.5	6	2,864,469	2,724,601	4,557,473	3,112,997	4,357,473	12%
		0.9	4	3,460,840	3,147,884	4,410,069	3,869,252	4,739,552	19%
	5	0	6	11,386,313	12,541,832	22,787,364	12,702,986	22,787,364	1%
		0.5	5	13,635,159	12,712,913	22,287,364	14,896,198	22,208,291	15%
		0.9	4	17,304,198	13,787,444	21,975,797	15,636,705	23,199,859	12%
	10	0	6	22,412,124	24,728,335	45,574,729	25,166,440	45,574,729	2%
		0.5	5	27,186,803	25,027,407	44,574,729	28,863,429	44,263,095	13%
		0.9	4	34,608,397	27,416,340	43,920,442	30,089,296	45,739,579	9%
200	1	0	8	2,599,840	2,662,764	4,757,473	2,748,503	4,657,473	3%
		0.5	6	3,054,078	2,712,652	4,557,473	2,980,460	4,457,473	9%
		0.9	4	4,415,710	3,084,216	4,357,473	3,727,040	4,610,719	17%
	5	0	6	12,225,772	12,463,377	22,787,364	12,722,661	22,787,364	2%
		0.5	5	14,870,099	12,674,534	22,287,364	13,031,319	22,287,364	3%
		0.9	3	22,045,780	14,222,971	21,617,923	14,901,907	22,265,378	5%
	10	0	6	24,168,288	24,654,132	45,574,729	25,128,612	45,574,729	2%
		0.5	5	29,646,948	24,957,044	44,574,729	25,611,150	44,574,729	3%
		0.9	3	44,013,689	27,629,179	43,180,752	28,889,111	44,519,917	4%

Q3. Why contextual ER-DRO, and are its conditions satisfied? The conditions were confirmed in Q1 (independence of the residuals from x and consistency of $\hat{f}$). The effect has two parts. *Effect of context*: in Table 7, the OOS expected-cost gain of the context model is 1–3% at $\zeta = 0$, 3–15% at $\zeta = 0.5$ and 4–19% at $\zeta = 0.9$. The gain is small because x has little explanatory power at $\beta_1 = 0.8$; in the β_1 panel of Figure 3, the two models have equal cost at $\beta_1 = 0$, and the gain grows to 1%, 16%, 25%, 41% at $\beta_1 = 0.8, 1.5, 3, 5$. *Effect of DRO*: ζ trades expected cost against tail cost. For $N = 200$, $c^- = 5$, as $\zeta = 0 \to 0.5 \to 0.9$ the expected cost is $12.46 \to 12.67 \to 14.22$M (+1.7%, +14%) and the tail cost is $22.79 \to 22.29 \to 21.62$M (−2.2%, −5.1%) (Figure 2). Meanwhile, the no-context model at $N = 15$, $\zeta = 0.9$ has a tail cost that is actually higher than at $\zeta = 0$ ($c^- = 5$: 22.79M→23.20M; likewise for $c^- = 1, 10$). This arises from trying to fit the tail with only 15 residuals and no x, and it means DRO should be used together with context.

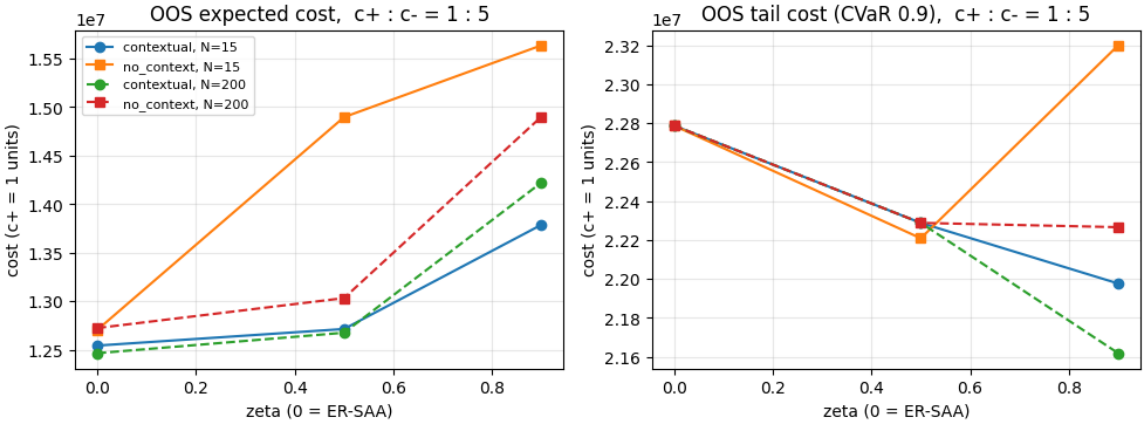


Figure 2: OOS expected cost (left) and tail cost (right) vs. ζ, $c^+ : c^- = 1 : 5$.

Q4. Can the model be used with scarce data? Yes, if context is used. In the $\zeta = 0.9$ rows of Table 7, the OOS expected-cost difference between $N = 15$ and $N = 200$ is 2.1%, 3.1%, 0.8% for $c^- = 1, 5, 10$, and the tail-cost difference is 1.2–1.7%. This holds even though $\hat{\beta}_1$ is overestimated by 39% at $N = 15$ and only the top 2 scenarios enter the objective at $\zeta = 0.9$. In the N panel of Figure 3, ER-SAA (context) is flat within 1.4% from $N = 10$ to 200, while ER-DRO (context) fluctuates over 13.19–14.24M and the no-context model over 14.38–16.14M. That is, the advantage of the context model grows as data become scarcer.

Q5. Sensitivity. Figure 3 changes one parameter at a time from the base setting ($N = 200$, $c^- = 5$, $\zeta = 0.9$). Cost rises with the entry cost F_0 and the variability σ_ε (F_0: 12.04→18.73M) and falls with the total fleet Q and the per-vehicle throughput κ (Q ratio 0.2→1.0: 16.79→11.45M). When Q is as small as 20% of expected demand the three models have almost the same cost, because every vehicle is used however it is allocated. The per-city cap $\bar{q}_c$ does not change the result above 100 for ER-SAA and above 150 for ER-DRO. In the variant (7) of Table 8, the second-stage cost falls as the number of entered cities grows over $B = 2, 3, 5$ ($N = 200$, $c^- = 5$: 14.57→12.73→11.74M), and the first three cities are stable regardless of N and c^-: Chicago, New York → San Diego. The composition at $B = 5$ depends on N; at $N = 200$, $c^- = 5, 10$, Philadelphia, Pittsburgh and Tampa enter instead of New York.

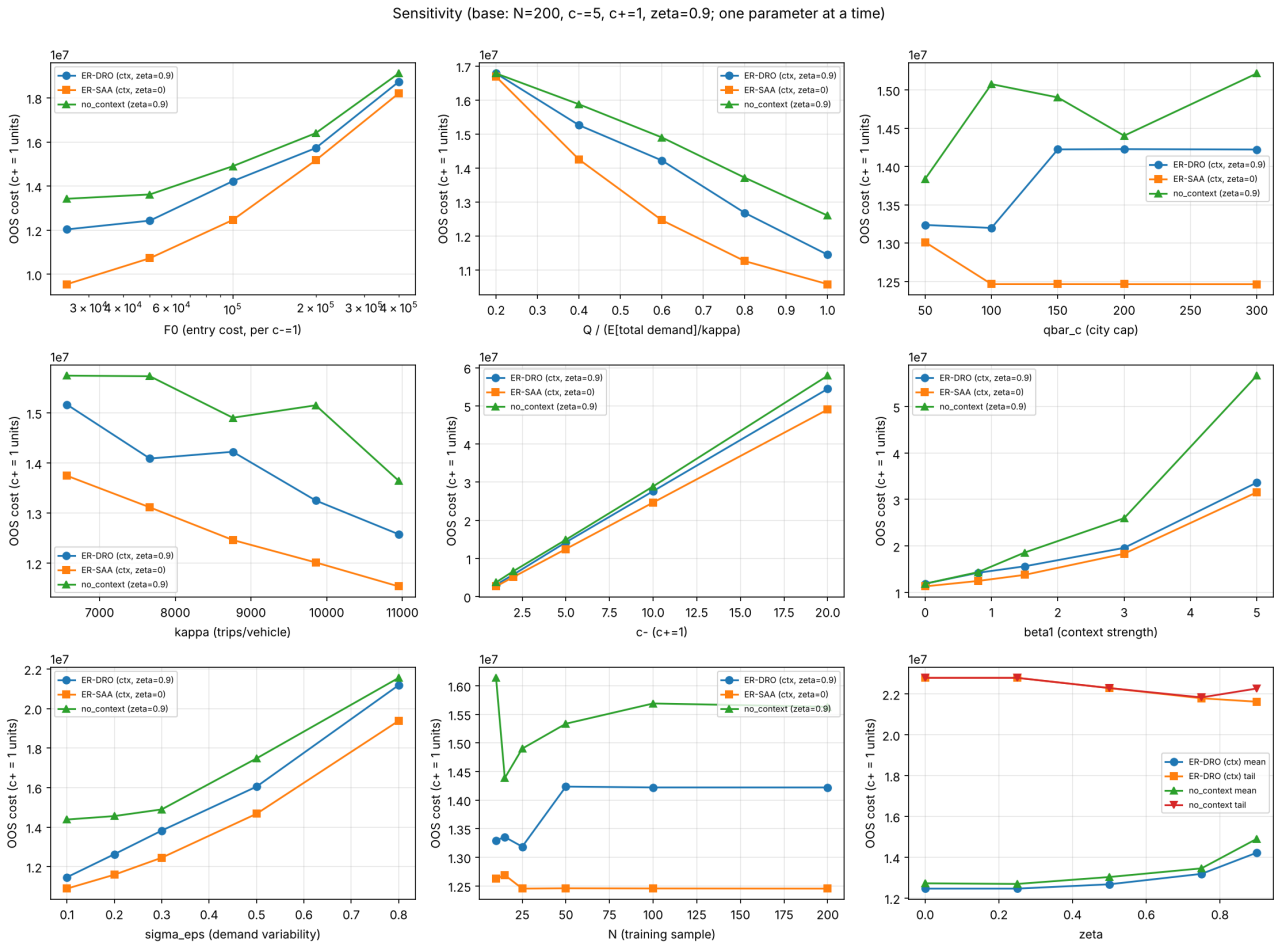


Figure 3: Sensitivity analysis. One parameter at a time from the base $N = 200$, $c^- = 5$, $c^+ = 1$, $\zeta = 0.9$. Lines: context ER-DRO ($\zeta = 0.9$), context ER-SAA ($\zeta = 0$), no context ($\zeta = 0.9$).

Table 8: Variant (7): OOS second-stage cost and selected cities by number of entered cities B ($\zeta = 0.9$)

N	c^-	B	OOS second-stage cost	Entered cities
15	1	2	3,315,579	Chicago, New York
		3	3,084,918	+ San Diego
		5	2,358,358	+ Philadelphia, Seattle
	5	2	14,339,887	Chicago, New York
		3	12,668,561	+ San Diego
		5	10,305,798	+ Philadelphia, Seattle
	10	2	28,111,826	Chicago, New York
		3	24,631,627	+ San Diego
		5	20,068,402	+ Philadelphia, Seattle
200	1	2	3,395,462	Chicago, New York
		3	3,198,004	+ San Diego
		5	2,440,092	+ Detroit, Philadelphia
	5	2	14,572,482	Chicago, New York
		3	12,726,413	+ San Diego
		5	11,735,447	Chicago, Philadelphia, Pittsburgh, San Diego, Tampa
	10	2	28,274,685	Chicago, New York
		3	24,625,772	+ San Diego
		5	22,821,183	Chicago, Philadelphia, Pittsburgh, San Diego, Tampa

4 Plan for real-data application (need to check data)

4.1 Data source by symbol

Table 9: Model inputs and real-data sources

Input	Real data	Source	Unit
p^k (training dependent variable)	robotaxi trips $\div M^k$	CPUC AV Passenger Service quarterly reports (submitted by operators; California)	tract × period
x^k (training context)	zero-vehicle household share	ACS 5-year estimates, B08201 (`_002E/_001E`)	tract
M^k (training)	$\sum_g$ (households with g vehicles)×(NHTS trip rate of group g)×days	ACS B08201 (`_002E`–`_006E`) + NHTS 2022	tract × period
x_{cj} (candidate-city tract)	zero-vehicle household share	ACS B08201, tracts within the candidate county	tract
M_{cj} (candidate-city tract)	same formula, annual	ACS + NHTS	tract × year
κ	8,760 trips/vehicle/year	back-calculated from Waymo's public figures	–
Q, $\bar{q}_c$, F_0	same assumptions as the toy	–	–
$c^+ : c^-$	1:1, 1:5, 1:10	no cost data → ratios	–
ζ	fixed at 0.9	–	–

4.2 Procedure

For the tracts $k = 1, \ldots, N$ of an operating city (e.g. San Francisco), observe x^k, M^k and the trip count T^k, and set $p^k = T^k/M^k$. For the tracts $j \in J_c$ of a candidate city c only x_{cj}, M_{cj} are available (no trips).

1. Regression (operating-city tracts).

$$\operatorname{logit}(p^k) = \beta_0 + \beta_1 x^k + \varepsilon^k, \qquad (\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{\beta} \sum_{k=1}^{N} \big(\operatorname{logit}(p^k) - \beta_0 - \beta_1 x^k\big)^2, \qquad \hat{\varepsilon}^k = \operatorname{logit}(p^k) - \hat{\beta}_0 - \hat{\beta}_1 x^k. \tag{10}$$

2. Scenarios (applied to candidate-city tracts; one residual = one scenario).

$$p^s_{cj} = \sigma\big(\hat{\beta}_0 + \hat{\beta}_1 x_{cj} + \hat{\varepsilon}^s\big), \qquad D^s_c = \sum_{j \in J_c} M_{cj}\, p^s_{cj}, \qquad s = 1, \ldots, N. \tag{11}$$

c is the candidate city and j a tract inside it (the unit at which demand is computed).

3. Optimization. Insert D^s_c into the extensive form (6) and solve for (y_c, q_c). The decision is at the city level; the demand is the sum over tracts.

4. Cross-city validation. The CPUC data include LA in addition to San Francisco. Estimate $\hat{f}$ from the San Francisco tracts, insert the x of the LA tracts, and compare the predicted p with the actual p of LA (MAE, R^2, whether the residual mean is 0, whether the spread matches the SF residuals). If the LA residual mean departs substantially from 0, the extrapolation has limits.

4.3 Assumptions

1. **Same relationship (extrapolation)**: β_0, β_1 estimated on operating-city tracts also hold for candidate-city tracts.
2. **Residual pooling**: the distribution of ε does not depend on the tract or on x (mean 0, homoscedastic), so that operating-city residuals can be attached to any candidate-city tract.

3. **Same unit**: estimation and application use the same spatial unit (tract), the numerator and denominator of p cover the same period, and M is computed by the same formula in both cities.
4. **Within range**: the candidate-tract x_{cj} lie within the range of the training x^k.
5. **Exogenous demand**: p does not depend on the decision q_c (decision-independent; relaxed in a later extension).

4.4 A small numerical example

Suppose San Francisco (operating) has only 3 tracts and Chicago (candidate) only 2. The period is one year.

SF tract	households	zero-vehicle households	x	M (households×10)	robotaxi trips (CPUC)	p = trips/M
A	1,000	100	0.10	10,000	300	0.03
B	1,000	300	0.30	10,000	600	0.06
C	1,000	500	0.50	10,000	800	0.08

Step 1. Training data (San Francisco tracts). x and the household counts come from ACS, the trip counts from CPUC. There are $N = 3$ observations. At the city level all of San Francisco would be a single row ($N = 1$) and regression would be impossible, which is why we go down to tracts.

Step 2. Regression. Fitting logit(p) on x gives $\hat{\beta}_0 = -3.67$, $\hat{\beta}_1 = 2.58$ and residuals $\hat{\varepsilon} = (-0.07, +0.14, -0.07)$. These three values become the three scenarios as they are.

Step 3. Application to the candidate city. Chicago has no trip data, only ACS.

Chicago tract	x	M
D	0.20	20,000
E	0.60	5,000

Scenario 1 (residual -0.07): tract D has $p = \sigma(-3.67 + 2.58 \times 0.20 - 0.07) = 0.0385$, tract E has $p = \sigma(-3.67 + 2.58 \times 0.60 - 0.07) = 0.1012$, and the demand is $D = 20{,}000(M_D) \times 0.0385(p_D) + 5{,}000(M_E) \times 0.1012(p_E) = 770 + 506 = 1{,}276$. Likewise scenario 2 (residual $+0.14$) gives $940 + 609 = 1{,}548$ and scenario 3 gives 1,276. These three values are the D^s_{Chicago}, $s = 1, 2, 3$ of the optimization. The decision (y, q) is made for the city as a whole; only the demand is computed as a sum over tracts.

Time unit. Even if the table above covered one month instead of one year, p = trips/M is on the same scale, so $\hat{f}$ can be used as is, and setting Chicago's M to an annual value gives annual demand. Monthly data, however, have more dispersed residuals, so the scenario variability is larger.